

CORE PURE / FURTHER CALCULUS

Volumes of Revolution

Worked solutions

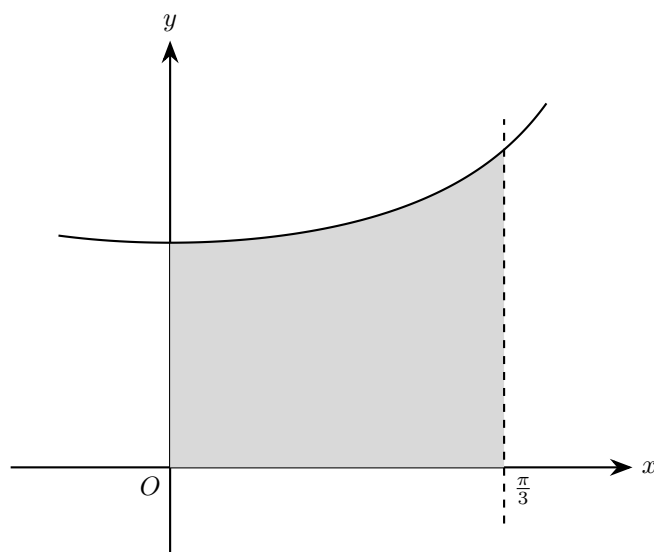
15 questions · 104 marks

Questions

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1. (a) Show that

$$\frac{d}{dx} \ln(\sec x + \tan x) = \sec x \quad [2]$$



The shaded region is enclosed by the curve $y = \sqrt{\sec x}$, the coordinate axes and the line $x = \frac{\pi}{3}$.

The region is rotated completely about the x -axis.

(b) Find the exact volume of the solid generated. [3]

Solution

(a) Let $u = \sec x + \tan x$. Differentiating,

$$\frac{du}{dx} = \sec x \tan x + \sec^2 x = \sec x (\tan x + \sec x) = u \sec x$$

Hence, by the chain rule,

$$\frac{d}{dx} \ln(\sec x + \tan x) = \frac{1}{u} \cdot \frac{du}{dx} = \frac{1}{u} \cdot u \sec x = \sec x$$

as required.

(b) Since the region is rotated about the x -axis, the volume is

$$V = \pi \int_a^b y^2 dx$$

Here the limits are $x = 0$ to $x = \frac{\pi}{3}$, and

$$y = \sqrt{\sec x} \implies y^2 = \sec x$$

So

$$V = \pi \int_0^{\pi/3} \sec x dx$$

From part (a),

$$\int \sec x dx = \ln |\sec x + \tan x| + c$$

so

$$V = \pi [\ln(\sec x + \tan x)]_0^{\pi/3}$$

Now substitute the limits:

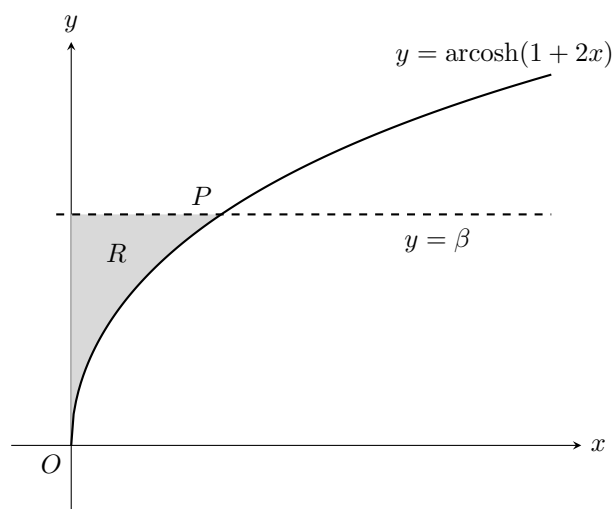
$$V = \pi \left(\ln \left(\sec \frac{\pi}{3} + \tan \frac{\pi}{3} \right) - \ln(\sec 0 + \tan 0) \right)$$

$$\sec \frac{\pi}{3} = 2, \quad \tan \frac{\pi}{3} = \sqrt{3}, \quad \sec 0 = 1, \quad \tan 0 = 0$$

Hence

$$V = \pi \left(\ln(2 + \sqrt{3}) - \ln 1 \right)$$

$$V = \pi \ln(2 + \sqrt{3})$$



2. The diagram above shows a sketch of part of the curve with equation

$$y = \operatorname{arcosh}(1 + 2x) \quad x \geq 0$$

and the straight line with equation $y = \beta$.

The line and the curve intersect at the point P with coordinates (α, β) .

Given that $\beta = \ln(2 + \sqrt{3})$,

(a) show that $\alpha = \frac{1}{2}$. [3]

The finite region R , shown shaded in the diagram above, is bounded by the curve with equation $y = \operatorname{arcosh}(1 + 2x)$, the y -axis and the line with equation $y = \beta$.

The region R is rotated through 2π radians about the y -axis.

(b) Use calculus to find the exact value of the volume of the solid generated. [6]

Solution

(a) At the point of intersection, (α, β) lies on the curve

$$y = \operatorname{arcosh}(1 + 2x)$$

so

$$\beta = \operatorname{arcosh}(1 + 2\alpha)$$

Applying cosh to both sides gives

$$\cosh \beta = 1 + 2\alpha$$

Given

$$\beta = \ln(2 + \sqrt{3})$$

and using

$$\operatorname{arcosh} 2 = \ln(2 + \sqrt{2^2 - 1}) = \ln(2 + \sqrt{3})$$

we have

$$\beta = \operatorname{arcosh} 2$$

so

$$\cosh \beta = 2$$

Therefore

$$1 + 2\alpha = 2$$

and hence

$$2\alpha = 1$$

so

$$\alpha = \frac{1}{2}$$

(b) First write x in terms of y .

From

$$y = \operatorname{arcosh}(1 + 2x)$$

we get

$$\cosh y = 1 + 2x$$

so

$$x = \frac{\cosh y - 1}{2}$$

When $x = 0$,

$$y = \operatorname{arcosh}(1) = 0$$

and the top of the region is $y = \beta$. Hence $0 \leq y \leq \beta$.

When the region is rotated about the y -axis, the volume is

$$V = \pi \int_0^\beta x^2 \, dy$$

So

$$V = \pi \int_0^\beta \left(\frac{\cosh y - 1}{2} \right)^2 \, dy = \frac{\pi}{4} \int_0^\beta (\cosh y - 1)^2 \, dy$$

Now expand:

$$(\cosh y - 1)^2 = \cosh^2 y - 2 \cosh y + 1$$

and use

$$\cosh^2 y = \frac{\cosh 2y + 1}{2}$$

to get

$$\begin{aligned} (\cosh y - 1)^2 &= \frac{\cosh 2y + 1}{2} - 2 \cosh y + 1 \\ &= \frac{1}{2} \cosh 2y - 2 \cosh y + \frac{3}{2} \end{aligned}$$

Therefore

$$\begin{aligned} V &= \frac{\pi}{4} \int_0^\beta \left(\frac{1}{2} \cosh 2y - 2 \cosh y + \frac{3}{2} \right) \, dy \\ &= \frac{\pi}{4} \left[\frac{1}{4} \sinh 2y - 2 \sinh y + \frac{3}{2} y \right]_0^\beta \end{aligned}$$

Now use the result from part (a), or equivalently $\beta = \operatorname{arcosh} 2$, so

$$\cosh \beta = 2$$

Hence

$$\sinh \beta = \sqrt{\cosh^2 \beta - 1} = \sqrt{4 - 1} = \sqrt{3}$$

since $\beta > 0$.

Also,

$$\sinh 2\beta = 2 \sinh \beta \cosh \beta = 2(\sqrt{3})(2) = 4\sqrt{3}$$

Substituting:

$$\begin{aligned} V &= \frac{\pi}{4} \left[\frac{1}{4}(4\sqrt{3}) - 2(\sqrt{3}) + \frac{3}{2}\beta \right] \\ &= \frac{\pi}{4} \left(\sqrt{3} - 2\sqrt{3} + \frac{3}{2}\beta \right) \\ &= \frac{\pi}{4} \left(-\sqrt{3} + \frac{3}{2}\beta \right) \\ &= \frac{\pi}{8} (3\beta - 2\sqrt{3}) \end{aligned}$$

Finally, since $\beta = \ln(2 + \sqrt{3})$,

$$V = \frac{\pi}{8} (3 \ln(2 + \sqrt{3}) - 2\sqrt{3})$$

3. The equation of a curve is

$$y = \sqrt{\frac{k-x}{k+x}}$$

where k is a positive constant. The region enclosed by the curve, the x -axis, the y -axis and the line $x = k$ is rotated through 2π radians about the x -axis.

Given that the volume of the solid of revolution formed is 1 unit^3 , find the exact value of k .

[4]

Solution

When the region is rotated about the x -axis, the volume is given by

$$V = \pi \int_0^k y^2 \, dx$$

Here

$$y = \sqrt{\frac{k-x}{k+x}}$$

so

$$y^2 = \frac{k-x}{k+x}$$

Given that the volume is 1 unit^3 ,

$$1 = \pi \int_0^k \frac{k-x}{k+x} \, dx$$

First simplify the integrand:

$$\frac{k-x}{k+x} = \frac{-(k+x) + 2k}{k+x} = -1 + \frac{2k}{k+x}$$

So

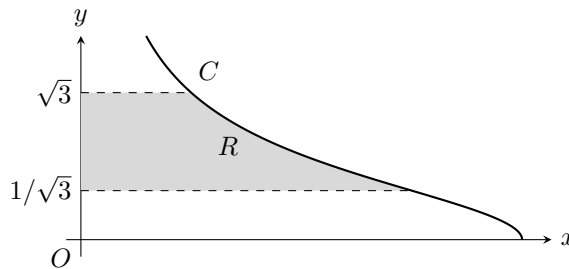
$$\begin{aligned} 1 &= \pi \int_0^k \left(-1 + \frac{2k}{k+x} \right) \, dx \\ &= \pi [-x + 2k \ln(k+x)]_0^k \\ &= \pi ([-k + 2k \ln(2k)] - [0 + 2k \ln k]) \\ &= \pi (-k + 2k(\ln(2k) - \ln k)) \\ &= \pi (-k + 2k \ln 2) \\ &= \pi k(2 \ln 2 - 1) \end{aligned}$$

Hence

$$k = \frac{1}{\pi(2 \ln 2 - 1)}$$

4. (a) Find

$$\int \sin^2 \theta \, d\theta \quad [2]$$



The diagram shows part of the curve C with parametric equations $x = 6 \sin^2 \theta$, $y = \cot \theta$, $0 < \theta \leq \frac{\pi}{2}$.

The finite region R shown in the diagram is bounded by C , the lines $y = \frac{1}{\sqrt{3}}$, $y = \sqrt{3}$ and the y -axis. Region R is rotated through 2π radians about the y -axis to form a solid of revolution.

(b) Show that the volume of the solid of revolution formed is given by the integral

$$k \int_a^b \sin^2 \theta \, d\theta$$

where a , b and k are constants to be found.

[5]

(c) Hence find the exact value of this volume, giving your answer in the form $p\pi^2$, where p is a constant to be found.

[3]

Solution

(a) Use the identity

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

Then

$$\begin{aligned} \int \sin^2 \theta \, d\theta &= \int \frac{1 - \cos 2\theta}{2} \, d\theta \\ &= \frac{1}{2} \int 1 \, d\theta - \frac{1}{2} \int \cos 2\theta \, d\theta \\ &= \frac{\theta}{2} - \frac{1}{2} \cdot \frac{\sin 2\theta}{2} + c \\ &= \frac{\theta}{2} - \frac{\sin 2\theta}{4} + c \end{aligned}$$

(b) When the region is rotated about the y -axis, the volume is

$$V = \pi \int x^2 \, dy$$

Here

$$x = 6 \sin^2 \theta, \quad y = \cot \theta$$

so

$$\frac{dy}{d\theta} = -\operatorname{cosec}^2 \theta$$

We now find the θ -limits.

When $y = \sqrt{3}$,

$$\cot \theta = \sqrt{3} \implies \tan \theta = \frac{1}{\sqrt{3}}$$

so $\theta = \frac{\pi}{6}$.

When $y = \frac{1}{\sqrt{3}}$,

$$\cot \theta = \frac{1}{\sqrt{3}} \implies \tan \theta = \sqrt{3}$$

so $\theta = \frac{\pi}{3}$.

Therefore

$$\begin{aligned} V &= \pi \int_{1/\sqrt{3}}^{\sqrt{3}} x^2 \, dy \\ &= \pi \int_{\pi/3}^{\pi/6} (6 \sin^2 \theta)^2 \frac{dy}{d\theta} \, d\theta \\ &= \pi \int_{\pi/3}^{\pi/6} (6 \sin^2 \theta)^2 (-\operatorname{cosec}^2 \theta) \, d\theta \\ &= 36\pi \int_{\pi/3}^{\pi/6} -\sin^2 \theta \, d\theta \\ &= 36\pi \int_{\pi/6}^{\pi/3} \sin^2 \theta \, d\theta \end{aligned}$$

So the required form is

$$k \int_a^b \sin^2 \theta \, d\theta$$

with

$$k = 36\pi, \quad a = \frac{\pi}{6}, \quad b = \frac{\pi}{3}$$

(c) Using parts (a) and (b),

$$\begin{aligned} V &= 36\pi \int_{\pi/6}^{\pi/3} \sin^2 \theta \, d\theta \\ &= 36\pi \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_{\pi/6}^{\pi/3} \\ &= 36\pi \left(\frac{\pi}{6} - \frac{\sin(2\pi/3)}{4} - \frac{\pi}{12} + \frac{\sin(\pi/3)}{4} \right) \end{aligned}$$

Since

$$\sin \frac{2\pi}{3} = \sin \frac{\pi}{3}$$

the sine terms cancel, giving

$$\begin{aligned} V &= 36\pi \left(\frac{\pi}{6} - \frac{\pi}{12} \right) \\ &= 36\pi \cdot \frac{\pi}{12} \\ &= 3\pi^2 \end{aligned}$$

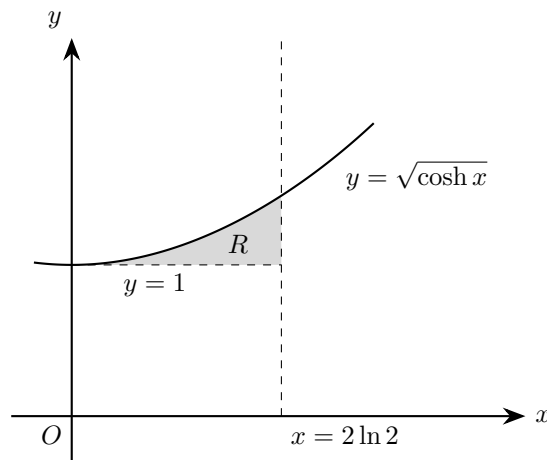
Hence the exact volume is

$$3\pi^2$$

so $p = 3$.

5. (a) Show that $\sinh(2 \ln 2) = \frac{15}{8}$

[2]



The region R is bounded by the curve with equation $y = \sqrt{\cosh x}$, the line $y = 1$ and the line $x = 2 \ln 2$, as shown in the diagram. The units of the axes are centimetres.

A designer models a hollow glass ornament as the solid formed by rotating R completely about the x -axis.

(b) Determine, according to the model, the exact volume of the ornament.

[4]

Solution

(a) Using

$$\sinh u = \frac{e^u - e^{-u}}{2}$$

with $u = 2 \ln 2$,

$$\begin{aligned} \sinh(2 \ln 2) &= \frac{e^{2 \ln 2} - e^{-2 \ln 2}}{2} \\ &= \frac{e^{\ln 4} - e^{-\ln 4}}{2} \\ &= \frac{4 - \frac{1}{4}}{2} \\ &= \frac{\frac{16}{4} - \frac{1}{4}}{2} = \frac{15/4}{2} = \frac{15}{8} \end{aligned}$$

Hence,

$$\sinh(2 \ln 2) = \frac{15}{8}$$

(b) First find the left-hand limit of the region by solving where the curve meets $y = 1$:

$$\begin{aligned} \sqrt{\cosh x} &= 1 \\ \cosh x &= 1 \end{aligned}$$

and this gives $x = 0$.

So the region extends from $x = 0$ to $x = 2 \ln 2$.

When the region is rotated about the x -axis, the volume is

$$V = \pi \int_a^b ((\text{upper curve})^2 - (\text{lower curve})^2) \, dx$$

Here, the upper curve is $y = \sqrt{\cosh x}$ and the lower curve is $y = 1$, so

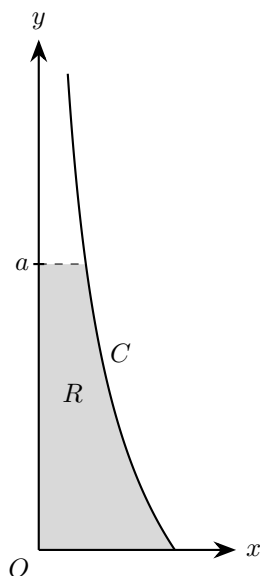
$$\begin{aligned} V &= \pi \int_0^{2 \ln 2} \left(\left(\sqrt{\cosh x} \right)^2 - 1^2 \right) dx \\ &= \pi \int_0^{2 \ln 2} (\cosh x - 1) dx \end{aligned}$$

Integrating,

$$\begin{aligned} V &= \pi [\sinh x - x]_0^{2 \ln 2} \\ &= \pi (\sinh(2 \ln 2) - 2 \ln 2 - (\sinh 0 - 0)) \\ &= \pi (\sinh(2 \ln 2) - 2 \ln 2) \end{aligned}$$

From part (a), $\sinh(2 \ln 2) = \frac{15}{8}$, so

$$V = \pi \left(\frac{15}{8} - 2 \ln 2 \right) \text{ cm}^3$$



6. The curve C is given parametrically by

$$x = \frac{9}{t^4}, \quad y = t^2 - 3, \quad t \geq \sqrt{3}$$

The shaded region R , shown in the diagram, is enclosed by C , the x -axis, the y -axis and the line $y = a$. When R is rotated through 2π radians about the y -axis, the resulting solid has volume

$$\frac{7\pi}{8}$$

Find the exact value of a .

[8]

Solution

First write the curve in the form x as a function of y .

From

$$y = t^2 - 3$$

we have

$$t^2 = y + 3$$

So

$$x = \frac{9}{t^4} = \frac{9}{(t^2)^2} = \frac{9}{(y+3)^2}$$

Since $t \geq \sqrt{3}$, we have $y \geq 0$, so the region runs from $y = 0$ up to $y = a$.

When the region is rotated about the y -axis, the volume is

$$V = \pi \int_0^a x^2 \, dy$$

Hence

$$\begin{aligned} V &= \pi \int_0^a \left(\frac{9}{(y+3)^2} \right)^2 dy \\ &= \pi \int_0^a \frac{81}{(y+3)^4} dy \\ &= 81\pi \int_0^a (y+3)^{-4} dy \\ &= 81\pi \left[\frac{(y+3)^{-3}}{-3} \right]_0^a \\ &= 81\pi \left[-\frac{1}{3(y+3)^3} \right]_0^a \\ &= 81\pi \left(-\frac{1}{3(a+3)^3} + \frac{1}{3 \cdot 3^3} \right) \\ &= 81\pi \left(-\frac{1}{3(a+3)^3} + \frac{1}{81} \right) \\ &= -\frac{27\pi}{(a+3)^3} + \pi \\ &= \pi \left(1 - \frac{27}{(a+3)^3} \right) \end{aligned}$$

We are told that the volume is $\frac{7\pi}{8}$, so

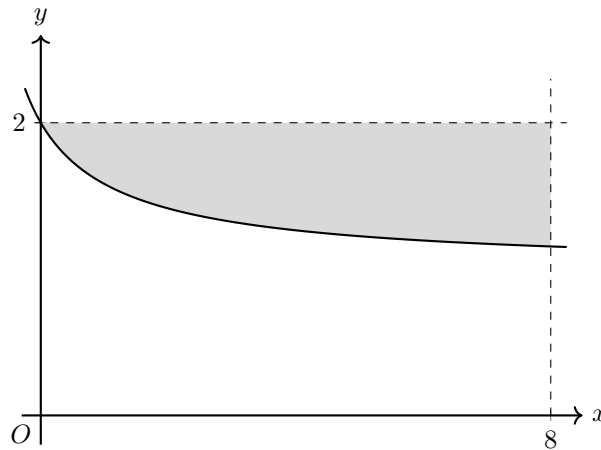
$$\begin{aligned} \pi \left(1 - \frac{27}{(a+3)^3} \right) &= \frac{7\pi}{8} \\ 1 - \frac{27}{(a+3)^3} &= \frac{7}{8} \\ \frac{27}{(a+3)^3} &= \frac{1}{8} \\ (a+3)^3 &= 216 \end{aligned}$$

Now $a \geq 0$, so $a+3 > 0$, and therefore

$$a+3 = 6$$

Thus

$$a = 3$$



7. The diagram shows the region bounded by the curve $y = \sqrt{\frac{x+4}{x+1}}$, the line $y = 2$ and the line $x = 8$. The curve meets the line $y = 2$ at the point $(0, 2)$.

Find the exact volume of the solid formed when this region is rotated through 360° about the x -axis. [6]

Solution

The region is between the line $y = 2$ and the curve

$$y = \sqrt{\frac{x+4}{x+1}}$$

from $x = 0$ to $x = 8$.

When this region is rotated about the x -axis, the volume is

$$V = \pi \int_0^8 \left(2^2 - \left(\sqrt{\frac{x+4}{x+1}} \right)^2 \right) dx$$

So

$$V = \pi \int_0^8 \left(4 - \frac{x+4}{x+1} \right) dx$$

Simplify the integrand:

$$\begin{aligned} 4 - \frac{x+4}{x+1} &= \frac{4(x+1) - (x+4)}{x+1} \\ &= \frac{4x+4-x-4}{x+1} \\ &= \frac{3x}{x+1} \end{aligned}$$

Hence

$$V = \pi \int_0^8 \frac{3x}{x+1} dx = 3\pi \int_0^8 \frac{x}{x+1} dx$$

Now write

$$\frac{x}{x+1} = 1 - \frac{1}{x+1}$$

Therefore

$$V = 3\pi \int_0^8 \left(1 - \frac{1}{x+1} \right) dx$$

Integrating gives

$$V = 3\pi [x - \ln(x+1)]_0^8$$

Substitute the limits:

$$\begin{aligned} V &= 3\pi ((8 - \ln 9) - (0 - \ln 1)) \\ &= 3\pi(8 - \ln 9) \end{aligned}$$

Since $\ln 9 = 2\ln 3$,

$$\begin{aligned} V &= 24\pi - 3\pi \ln 9 \\ &= 24\pi - 6\pi \ln 3 \\ &= 6\pi(4 - \ln 3) \end{aligned}$$

8. The region in the first quadrant bounded by the curve $y = \sinh x$, the y -axis, and the line $y = 2$ is rotated through 360° about the x -axis.

Find the exact volume of revolution generated, expressing your answer in a form involving a logarithm.

[7]

Solution

Let the curve $y = \sinh x$ meet the line $y = 2$ at $x = a$.

Then

$$\sinh a = 2$$

To find a in logarithmic form,

$$\begin{aligned}\sinh a = 2 &\implies \frac{e^a - e^{-a}}{2} = 2 \\ &\implies e^a - e^{-a} = 4 \\ &\implies e^{2a} - 1 = 4e^a\end{aligned}$$

Let $u = e^a$. Then $u > 0$ and

$$u^2 - 4u - 1 = 0$$

So

$$u = \frac{4 \pm \sqrt{16 + 4}}{2} = 2 \pm \sqrt{5}$$

Since $u > 0$, we take

$$u = 2 + \sqrt{5}$$

Hence

$$a = \ln(2 + \sqrt{5})$$

For $0 \leq x \leq a$, the region lies between $y = \sinh x$ and $y = 2$. When it is rotated about the x -axis, the volume is

$$V = \pi \int_0^a (2^2 - (\sinh x)^2) \, dx$$

So

$$V = \pi \int_0^a (4 - \sinh^2 x) \, dx$$

Use

$$\sinh^2 x = \frac{\cosh 2x - 1}{2}$$

Then

$$\begin{aligned}V &= \pi \int_0^a \left(4 - \frac{\cosh 2x - 1}{2} \right) \, dx \\ &= \pi \int_0^a \left(\frac{9}{2} - \frac{1}{2} \cosh 2x \right) \, dx \\ &= \pi \left[\frac{9}{2}x - \frac{1}{4} \sinh 2x \right]_0^a\end{aligned}$$

Now $\sinh a = 2$, and since $a > 0$,

$$\cosh a = \sqrt{1 + \sinh^2 a} = \sqrt{1 + 4} = \sqrt{5}$$

Therefore

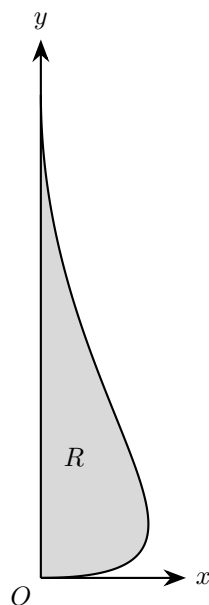
$$\sinh 2a = 2 \sinh a \cosh a = 2(2)(\sqrt{5}) = 4\sqrt{5}$$

Substituting into the volume,

$$\begin{aligned}V &= \pi \left(\frac{9}{2}a - \frac{1}{4} \sinh 2a \right) \\ &= \pi \left(\frac{9}{2}a - \frac{1}{4}(4\sqrt{5}) \right) \\ &= \pi \left(\frac{9}{2}a - \sqrt{5} \right)\end{aligned}$$

Finally, using $a = \ln(2 + \sqrt{5})$,

$$V = \pi \left(\frac{9}{2} \ln(2 + \sqrt{5}) - \sqrt{5} \right)$$



9. Part of the side profile of a solid wooden ornament is shown in the diagram.
The outline is modelled by the curve with parametric equations

$$x = 12t(1-t)^2, \quad y = 8t^2, \quad 0 \leq t \leq 1$$

The ornament is formed by rotating the shaded region through 2π radians about the y -axis.

Use the model to find the volume of the ornament.

[7]

Solution

For a solid formed by rotating a region about the y -axis, we use

$$V = \pi \int x^2 dy$$

Here

$$x = 12t(1-t)^2, \quad y = 8t^2, \quad 0 \leq t \leq 1$$

Also,

$$\frac{dy}{dt} = 16t$$

Since $0 \leq t \leq 1$, we have $\frac{dy}{dt} \geq 0$, so y increases through the interval and we can write

$$\begin{aligned} V &= \pi \int_0^1 x^2 \frac{dy}{dt} dt \\ &= \pi \int_0^1 (12t(1-t)^2)^2 (16t) dt \\ &= \pi \int_0^1 144t^2(1-t)^4 \cdot 16t dt \\ &= 2304\pi \int_0^1 t^3(1-t)^4 dt \end{aligned}$$

Now expand

$$(1-t)^4 = 1 - 4t + 6t^2 - 4t^3 + t^4$$

So

$$t^3(1-t)^4 = t^3 - 4t^4 + 6t^5 - 4t^6 + t^7$$

Hence

$$\begin{aligned} V &= 2304\pi \int_0^1 (t^3 - 4t^4 + 6t^5 - 4t^6 + t^7) \, dt \\ &= 2304\pi \left[\frac{t^4}{4} - \frac{4t^5}{5} + t^6 - \frac{4t^7}{7} + \frac{t^8}{8} \right]_0^1 \\ &= 2304\pi \left(\frac{1}{4} - \frac{4}{5} + 1 - \frac{4}{7} + \frac{1}{8} \right) \end{aligned}$$

Using a common denominator of 280,

$$\frac{1}{4} - \frac{4}{5} + 1 - \frac{4}{7} + \frac{1}{8} = \frac{70 - 224 + 280 - 160 + 35}{280} = \frac{1}{280}$$

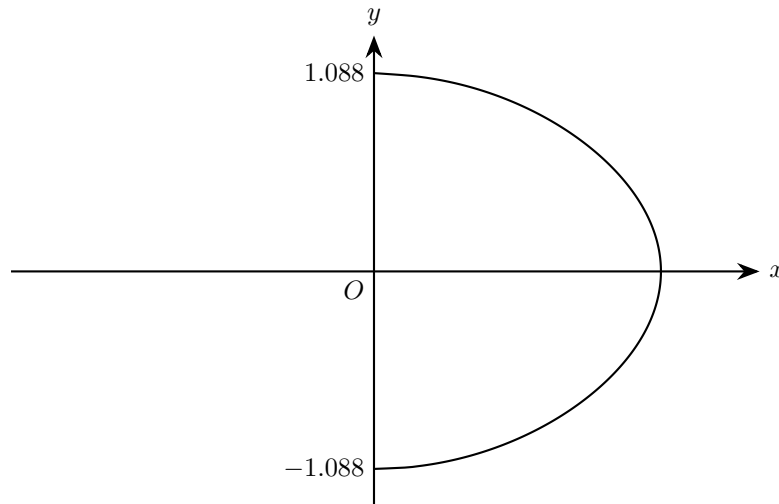
Therefore

$$\begin{aligned} V &= 2304\pi \cdot \frac{1}{280} \\ &= \frac{288\pi}{35} \end{aligned}$$

So the volume of the ornament is

$$\frac{288\pi}{35} \text{ units}^3$$

which is approximately 25.9 units³.



10. A chocolatier makes hand-rolled chocolate drops.

The tallest drop in one tray was approximately 2.2 cm high.

The shape of this drop is modelled by rotating the curve with equation

$$20x^2 + 6y^2 + y \sin(2y) = 8, \quad x \geq 0$$

shown in the diagram above, about the y -axis through 2π radians, where the units are cm.

Given that the y -intercepts of the curve are -1.088 and 1.088 to four significant figures,

- (a) Use algebraic integration to determine, according to the model, the volume of this chocolate drop. [6]

The chocolatier melts down 80 chocolate drops from the same tray.

- (b) Use your answer to part (a) to decide whether, in reality, there is likely to be enough chocolate to fill a mould of volume 140 cm^3 , giving a reason. [2]

Solution

- (a) At the y -intercepts, $x = 0$, so the limits of integration are

$$y = -1.088 \quad \text{and} \quad y = 1.088$$

From

$$20x^2 + 6y^2 + y \sin(2y) = 8$$

we rearrange to get

$$x^2 = \frac{8 - 6y^2 - y \sin(2y)}{20}$$

When this curve is rotated about the y -axis, the volume is

$$V = \pi \int_{-1.088}^{1.088} x^2 \, dy$$

So

$$V = \frac{\pi}{20} \int_{-1.088}^{1.088} (8 - 6y^2 - y \sin(2y)) \, dy$$

Using integration by parts for $\int -y \sin(2y) \, dy$, let

$$u = y, \quad \frac{dv}{dy} = -\sin(2y)$$

Then

$$\frac{du}{dy} = 1, \quad v = \frac{1}{2} \cos(2y)$$

Hence

$$\begin{aligned} \int -y \sin(2y) \, dy &= uv - \int v \frac{du}{dy} \, dy \\ &= \frac{y}{2} \cos(2y) - \int \frac{1}{2} \cos(2y) \, dy \\ &= \frac{y}{2} \cos(2y) - \frac{1}{4} \sin(2y) \end{aligned}$$

Therefore

$$\begin{aligned} V &= \frac{\pi}{20} \left[8y - 2y^3 + \frac{y}{2} \cos(2y) - \frac{1}{4} \sin(2y) \right]_{-1.088}^{1.088} \\ &\approx 1.763 \dots \end{aligned}$$

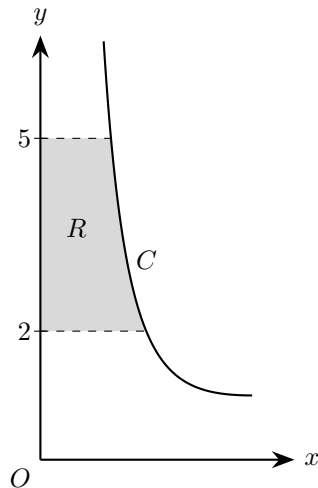
Hence, according to the model, the volume of the chocolate drop is 1.76 cm^3 .

(b) Using the answer from part (a), the volume of 80 such drops would be estimated as

$$80 \times 1.76 = 140.8 \text{ cm}^3$$

This is only just above 140 cm^3 . Also, the model in part (a) is for the tallest drop in the tray, so using 1.76 cm^3 for all 80 drops will overestimate the total amount of chocolate.

Therefore, in reality, there is unlikely to be enough chocolate to fill the 140 cm^3 mould.



11. The diagram shows part of the curve C with parametric equations

$$x = \frac{2}{1+t}, \quad y = t^2 + 1, \quad t \geq 0$$

The region R is bounded by the curve, the y -axis and the lines $y = 2$ and $y = 5$. Region R is rotated through 2π radians about the y -axis.

Use parametric integration to find the volume of the resulting solid of revolution.

[6]

Solution

For the given curve,

$$x = \frac{2}{1+t}, \quad y = t^2 + 1, \quad t \geq 0$$

First find the values of t corresponding to the horizontal boundaries.

When $y = 2$,

$$t^2 + 1 = 2 \implies t^2 = 1 \implies t = 1$$

since $t \geq 0$.

When $y = 5$,

$$t^2 + 1 = 5 \implies t^2 = 4 \implies t = 2$$

So the required part of the curve corresponds to $1 \leq t \leq 2$.

When the region is rotated about the y -axis, the volume is

$$V = \pi \int x^2 dy$$

Using the parameter t ,

$$V = \pi \int_1^2 x^2 \frac{dy}{dt} dt$$

Now

$$x^2 = \left(\frac{2}{1+t} \right)^2 = \frac{4}{(1+t)^2}$$

and

$$\frac{dy}{dt} = 2t$$

Hence

$$V = \pi \int_1^2 \frac{4}{(1+t)^2} \cdot 2t dt = 8\pi \int_1^2 \frac{t}{(1+t)^2} dt$$

Now simplify the integrand:

$$\frac{t}{(1+t)^2} = \frac{(1+t)-1}{(1+t)^2} = \frac{1}{1+t} - \frac{1}{(1+t)^2}$$

So

$$V = 8\pi \int_1^2 \left(\frac{1}{1+t} - \frac{1}{(1+t)^2} \right) dt$$

Integrating,

$$\int \frac{1}{1+t} dt = \ln(1+t)$$

and

$$\int -\frac{1}{(1+t)^2} dt = \frac{1}{1+t}$$

Therefore

$$V = 8\pi \left[\ln(1+t) + \frac{1}{1+t} \right]_1^2$$

Substituting the limits,

$$V = 8\pi \left(\ln 3 + \frac{1}{3} - \ln 2 - \frac{1}{2} \right)$$

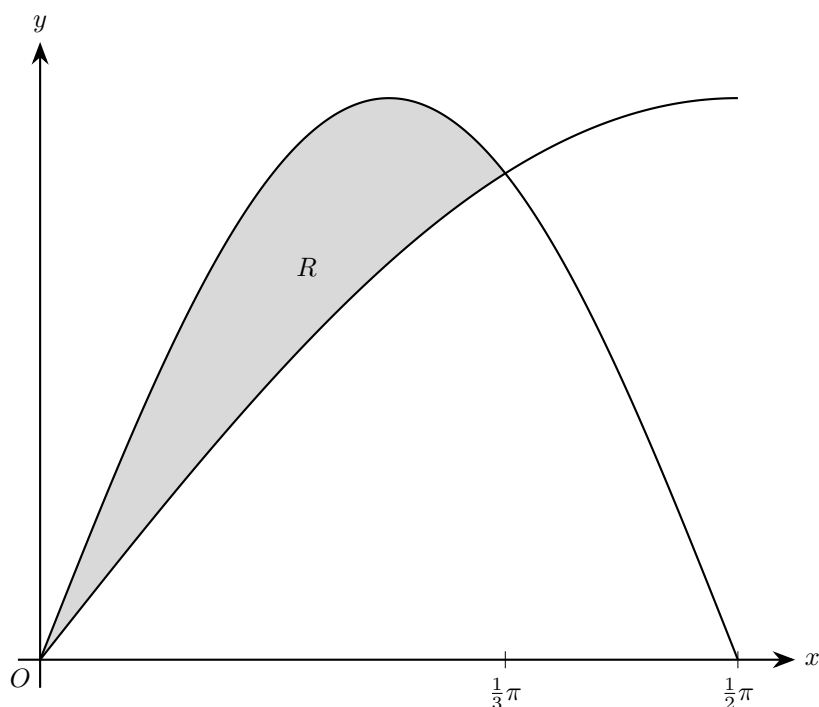
So

$$V = 8\pi \left(\ln \frac{3}{2} - \frac{1}{6} \right)$$

$$V = 8\pi \ln \frac{3}{2} - \frac{4\pi}{3}$$

In decimal form,

$$V \approx 6.00$$



12. The diagram shows the curves $y = \sin x$ and $y = \sin 2x$, for $0 \leq x \leq \frac{1}{2}\pi$. The shaded region R is the finite region enclosed by the curves.

Find the volume of the solid of revolution formed when R is rotated completely about the x -axis, giving your answer in terms of π .

[7]

Solution

First find where the curves meet.

$$\sin x = \sin 2x$$

Using $\sin 2x = 2 \sin x \cos x$,

$$\sin x = 2 \sin x \cos x$$

$$\sin x(2 \cos x - 1) = 0$$

So, for $0 \leq x \leq \frac{\pi}{2}$,

$$\sin x = 0 \Rightarrow x = 0 \quad \text{or} \quad 2 \cos x - 1 = 0 \Rightarrow \cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}$$

Hence the shaded region is between $x = 0$ and $x = \frac{\pi}{3}$.

For $0 < x < \frac{\pi}{3}$, we have $\sin x > 0$ and $\cos x > \frac{1}{2}$, so $2 \cos x - 1 > 0$. Therefore

$$\sin x(2 \cos x - 1) > 0$$

which means

$$\sin 2x > \sin x$$

So when the region is rotated about the x -axis, the volume is

$$V = \pi \int_0^{\pi/3} ((\sin 2x)^2 - (\sin x)^2) \, dx$$

Now use

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

Then

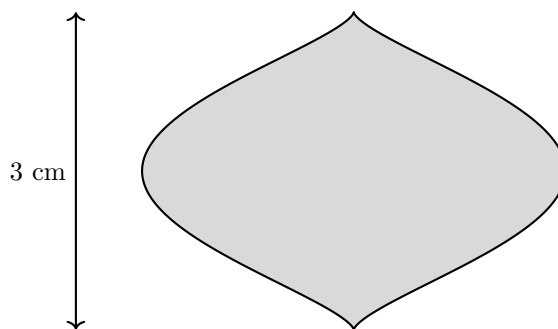
$$\begin{aligned}\sin^2 2x - \sin^2 x &= \frac{1 - \cos 4x}{2} - \frac{1 - \cos 2x}{2} \\ &= \frac{\cos 2x - \cos 4x}{2}\end{aligned}$$

So

$$\begin{aligned}V &= \pi \int_0^{\pi/3} \frac{\cos 2x - \cos 4x}{2} \mathrm{d}x \\ &= \pi \left[\frac{\sin 2x}{4} - \frac{\sin 4x}{8} \right]_0^{\pi/3}\end{aligned}$$

Now substitute the limits:

$$\begin{aligned}V &= \pi \left(\frac{\sin(2\pi/3)}{4} - \frac{\sin(4\pi/3)}{8} \right) \\ &= \pi \left(\frac{\sqrt{3}/2}{4} - \frac{-\sqrt{3}/2}{8} \right) \\ &= \pi \left(\frac{\sqrt{3}}{8} + \frac{\sqrt{3}}{16} \right) \\ &= \frac{3\sqrt{3}\pi}{16}\end{aligned}$$



13. The diagram shows the outline of a glass bead. The bead is modelled by the solid obtained when the region enclosed by a curve C is rotated about the y -axis. The actual bead has height 3 cm. The curve C has parametric equations

$$x = \sin \theta - \frac{1}{3} \sin 3\theta, \quad y = 1 - \cos \theta, \quad 0 \leq \theta \leq 2\pi$$

- (a) Show that a Cartesian equation of the curve C is

$$x^2 = \frac{16}{9}y^3(2-y)^3 \quad [4]$$

- (b) Hence, using the model, find, in cm^3 , the volume of the bead. [5]

Solution

- (a) Using the identity

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$

we have

$$\begin{aligned} x &= \sin \theta - \frac{1}{3} \sin 3\theta \\ &= \sin \theta - \frac{1}{3} (3 \sin \theta - 4 \sin^3 \theta) \\ &= \sin \theta - \sin \theta + \frac{4}{3} \sin^3 \theta \\ &= \frac{4}{3} \sin^3 \theta \end{aligned}$$

So

$$x^2 = \frac{16}{9} \sin^6 \theta$$

Also

$$y = 1 - \cos \theta$$

so

$$\cos \theta = 1 - y$$

Hence

$$\begin{aligned} \sin^2 \theta &= 1 - \cos^2 \theta \\ &= 1 - (1 - y)^2 \\ &= 1 - (1 - 2y + y^2) \\ &= 2y - y^2 \\ &= y(2 - y) \end{aligned}$$

Therefore

$$\sin^6 \theta = (\sin^2 \theta)^3 = (y(2 - y))^3$$

and so

$$\begin{aligned} x^2 &= \frac{16}{9} \sin^6 \theta \\ &= \frac{16}{9} (y(2 - y))^3 \\ &= \frac{16}{9} y^3 (2 - y)^3 \end{aligned}$$

Thus a Cartesian equation of C is

$$x^2 = \frac{16}{9} y^3 (2 - y)^3$$

(b) The model has height

$$2 - 0 = 2$$

whereas the actual bead has height 3 cm, so the linear scale factor is

$$k = \frac{3}{2}$$

First find the volume of the model.

When the region is rotated about the y -axis,

$$V_{\text{model}} = \pi \int_0^2 x^2 \, dy$$

Using the result from part (a),

$$\begin{aligned} V_{\text{model}} &= \pi \int_0^2 \frac{16}{9} y^3 (2 - y)^3 \, dy \\ &= \frac{16\pi}{9} \int_0^2 y^3 (2 - y)^3 \, dy \end{aligned}$$

Now expand:

$$\begin{aligned} (2 - y)^3 &= 8 - 12y + 6y^2 - y^3 \\ y^3(2 - y)^3 &= 8y^3 - 12y^4 + 6y^5 - y^6 \end{aligned}$$

So

$$\begin{aligned} V_{\text{model}} &= \frac{16\pi}{9} \int_0^2 (8y^3 - 12y^4 + 6y^5 - y^6) \, dy \\ &= \frac{16\pi}{9} \left[2y^4 - \frac{12}{5}y^5 + y^6 - \frac{1}{7}y^7 \right]_0^2 \end{aligned}$$

Substituting the limits,

$$\begin{aligned} V_{\text{model}} &= \frac{16\pi}{9} \left(2(2^4) - \frac{12}{5}(2^5) + 2^6 - \frac{1}{7}(2^7) \right) \\ &= \frac{16\pi}{9} \left(32 - \frac{384}{5} + 64 - \frac{128}{7} \right) \\ &= \frac{16\pi}{9} \cdot \frac{32}{35} \\ &= \frac{512\pi}{315} \end{aligned}$$

Volume scales as the cube of the linear scale factor, so

$$\begin{aligned} V_{\text{actual}} &= \left(\frac{3}{2}\right)^3 V_{\text{model}} \\ &= \frac{27}{8} \times \frac{512\pi}{315} \\ &= \frac{192\pi}{35} \end{aligned}$$

Therefore the volume of the bead is

$$\frac{192\pi}{35} \text{ cm}^3 \approx 17.2 \text{ cm}^3$$

14. The equation of a curve is

$$y = \frac{x}{\sqrt{k^2 + x^2}}$$

where k is a positive constant.

The region in the first quadrant bounded by the curve, the x -axis, the y -axis and the line $x = k$ is rotated through 2π radians about the x -axis.

Given that the volume of revolution formed is 1 unit³, find the exact value of k .

[4]

Solution

When the region is rotated about the x -axis, the volume is

$$V = \pi \int_0^k y^2 \, dx$$

Here

$$y = \frac{x}{\sqrt{k^2 + x^2}} \quad \Rightarrow \quad y^2 = \frac{x^2}{k^2 + x^2}$$

So

$$V = \pi \int_0^k \frac{x^2}{k^2 + x^2} \, dx$$

Rewrite the integrand:

$$\frac{x^2}{k^2 + x^2} = \frac{k^2 + x^2 - k^2}{k^2 + x^2} = 1 - \frac{k^2}{k^2 + x^2}$$

Hence

$$V = \pi \int_0^k \left(1 - \frac{k^2}{k^2 + x^2} \right) \, dx$$

Now integrate:

$$\begin{aligned} \int \left(1 - \frac{k^2}{k^2 + x^2} \right) \, dx &= x - k^2 \int \frac{1}{k^2 + x^2} \, dx \\ &= x - k^2 \left(\frac{1}{k} \arctan \frac{x}{k} \right) \\ &= x - k \arctan \frac{x}{k} \end{aligned}$$

Therefore

$$\begin{aligned} V &= \pi \left[x - k \arctan \frac{x}{k} \right]_0^k \\ &= \pi (k - k \arctan 1) \\ &= \pi \left(k - k \cdot \frac{\pi}{4} \right) \\ &= \pi k \left(1 - \frac{\pi}{4} \right) \end{aligned}$$

We are given that the volume is 1, so

$$\pi k \left(1 - \frac{\pi}{4} \right) = 1$$

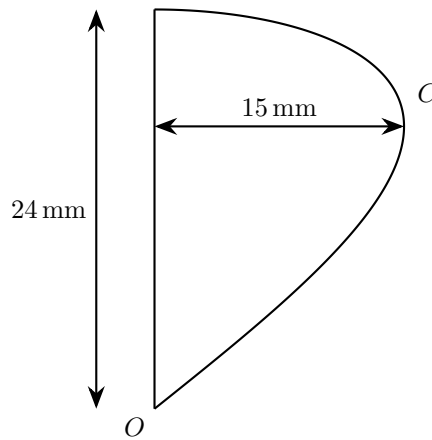
Solving for k :

$$\begin{aligned} k &= \frac{1}{\pi \left(1 - \frac{\pi}{4} \right)} \\ &= \frac{1}{\pi \left(\frac{4 - \pi}{4} \right)} \\ &= \frac{4}{\pi(4 - \pi)} \end{aligned}$$

Since k is positive, this is valid.

15. (a) Prove that

$$\cos 4\theta \cos \theta \equiv \frac{1}{2}(\cos 5\theta + \cos 3\theta) \quad [3]$$



The diagram shows the cross-section of a decorative ornament.

The ornament can be modelled by rotating the region enclosed by the curve C and the y -axis about the y -axis. Curve C has parametric equations

$$x = 15 \sin 2\theta, \quad y = 24 \sin \theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

where x and y are measured in millimetres.

(b) Find the volume of the ornament. [5]

Solution

(a) Using the cosine addition formula,

$$\cos 5\theta = \cos(4\theta + \theta) = \cos 4\theta \cos \theta - \sin 4\theta \sin \theta$$

and

$$\cos 3\theta = \cos(4\theta - \theta) = \cos 4\theta \cos \theta + \sin 4\theta \sin \theta$$

Adding these gives

$$\begin{aligned} \cos 5\theta + \cos 3\theta &= (\cos 4\theta \cos \theta - \sin 4\theta \sin \theta) + (\cos 4\theta \cos \theta + \sin 4\theta \sin \theta) \\ &= 2 \cos 4\theta \cos \theta \end{aligned}$$

Therefore

$$\cos 4\theta \cos \theta \equiv \frac{1}{2}(\cos 5\theta + \cos 3\theta)$$

So the required identity is proved.

(b) When the region is rotated about the y -axis, the volume is

$$V = \pi \int x^2 dy$$

Using the parameter θ ,

$$V = \pi \int_0^{\pi/2} x^2 \frac{dy}{d\theta} d\theta$$

Now

$$x = 15 \sin 2\theta, \quad y = 24 \sin \theta$$

so

$$\frac{dy}{d\theta} = 24 \cos \theta$$

Substituting,

$$\begin{aligned} V &= \pi \int_0^{\pi/2} (15 \sin 2\theta)^2 (24 \cos \theta) d\theta \\ &= 5400\pi \int_0^{\pi/2} \sin^2 2\theta \cos \theta d\theta \end{aligned}$$

Use

$$\sin^2 2\theta = \frac{1 - \cos 4\theta}{2}$$

Then

$$\begin{aligned} V &= 5400\pi \int_0^{\pi/2} \frac{1 - \cos 4\theta}{2} \cos \theta d\theta \\ &= 2700\pi \int_0^{\pi/2} (1 - \cos 4\theta) \cos \theta d\theta \\ &= 2700\pi \left(\int_0^{\pi/2} \cos \theta d\theta - \int_0^{\pi/2} \cos 4\theta \cos \theta d\theta \right) \end{aligned}$$

From part (a),

$$\cos 4\theta \cos \theta = \frac{1}{2}(\cos 5\theta + \cos 3\theta)$$

So

$$\begin{aligned} V &= 2700\pi \int_0^{\pi/2} \left(\cos \theta - \frac{1}{2}(\cos 5\theta + \cos 3\theta) \right) d\theta \\ &= 2700\pi \left[\sin \theta - \frac{1}{10} \sin 5\theta - \frac{1}{6} \sin 3\theta \right]_0^{\pi/2} \end{aligned}$$

Now evaluate the bracket:

$$\sin \frac{\pi}{2} = 1, \quad \sin \frac{5\pi}{2} = -1, \quad \sin \frac{3\pi}{2} = -1$$

Hence

$$\begin{aligned} V &= 2700\pi \left(1 - \frac{1}{10} - \frac{1}{6}(-1) \right) \\ &= 2700\pi \left(1 - \frac{1}{10} + \frac{1}{6} \right) \\ &= 2700\pi \left(\frac{16}{15} \right) \\ &= 2880\pi \end{aligned}$$

Therefore the volume of the ornament is

$$2880\pi \text{ mm}^3$$

which is approximately

$$9.05 \times 10^3 \text{ mm}^3$$